Grid generation on free-form surface using guide line advancing and surface flattening method

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Abstract: Automatic grid generation on a curved surface is important to more efficient design of free-form structures. However, it is neither a convenient nor an obvious task for engineers to create a discrete grid structure on a complex surface that meets the architectural requirements. Besides, research on the rapid grid generation methodology for free form structural design is still limited. In order to achieve better grid distribution of rods on free-form surface, a grid generation methodology which combines surface flattening technique with guide line method is put forward. The parametric domain of the free-form surface was firstly divided into a number of parts and a discrete free-form surface was accordingly formed by mapping the generated dividing points onto the curved surface. The free-form surface was then flattened based on the principle of identical area. Accordingly, the flattened rectangular lattices were fitted into the 2D surface where grids were formed by using the guide line method. Finally, the 2D grids were mapped onto the 3D surface, and grids were therefore generated on the given surface. A grid shape and rod length quality index was proposed to evaluate the shape of grid cells and rod lengths. The results show that the grid shape quality index and the deviation of rod length of the grid structure are reduced by up to 47% and 34% respectively by using the guide line method with surface flattening when compared to the method without surface flattening.

Keywords: free-form surface; grid generation; surface flattening; guide line method; Grid quality index

1 Introductions

In recent years, traditional spatial structural shapes [1-3] including plane, cylindrical surface, spherical surface, and parabolic surface have been widely used around the world. However, these traditional shapes are becoming more difficult to meet people's aesthetic requirements of variation in terms of architectural appearance. At the same time, parametric modeling and scripting techniques in computer aided design have enabled a new level of sophistication in 3D free-form surface, allowing engineers searching for ways to restructure the design process. Structural forms are increasing due to the expression of architectural creativity and the application of modern technology to building design. Complex free-form grid structure is one of the most striking trends in contemporary architecture [4, 5]. Free-form surface has the characteristic that is unable to be expressed accurately by means of one or several analytic functions and the curvature of such a surface is in a complex shape, as shown in Fig.1.

(a) Yas Viceroy Abu Dhabi Hotel

(b) Shenzhen Bao'an International Airport

Fig.1 Free-form grid structures

If an appropriate curved surface of such a form is selected and well-positioned supports are added, an efficient flow of forces within will be obtained [6]. This structural form is also more desirable due to the reductions in material usage and increase in service space.

However, it is not always obvious and convenient to create an efficient grid structure on a given surface. Fig.2 illustrates an assembly process of a structure, Wuhan Railway Station (Fig.2(c)). The architect gave the curved free-form surface (as shown in Fig.2(a)) in the early concept design stage, however, the structural analysis and assembly required an arrangement of nodes and elements on the curved surface based on experience and was therefore conducted manually with a tedious iteration, as shown in Fig.2(b). With the increasing application of free-form grid structures, a practical grid generation framework that can generate quick structural grids on a given free-form surface, will be necessary, especially in the early design stage.

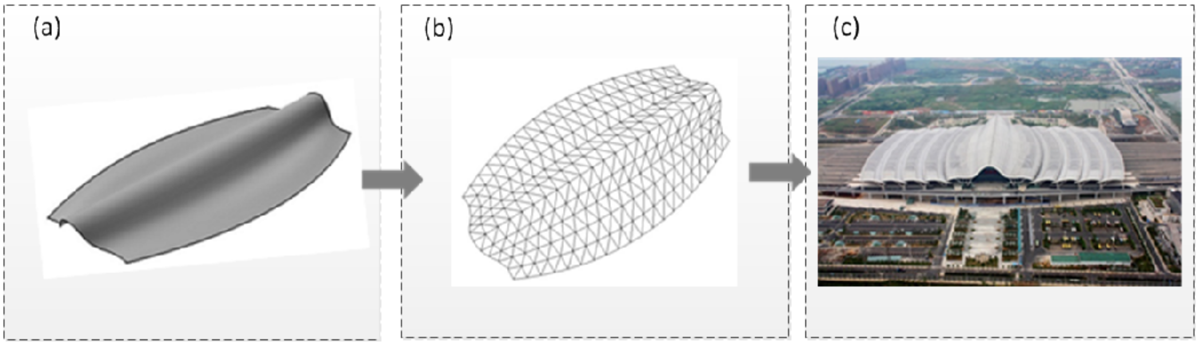


Fig.2 Structural assembling process of a free-form single layer structure (Wuhan Railway Station, Wuhan City, China)

Previous research on grid generation methodologies over the free-form surface is still limited. A promising method to create an efficient structural grid on an imposed surface is by topology optimisation, a good trial for this method was the research by Peter et al. [7] and Paul and Will [8]. The topology optimization was proved to be efficient for grid generation. However, as pointed by Peter et al. [7], the resulted topology were inapplicable directly, refinement was therefore required.

In order to obtain a usable and optimum grid over a free-form surface considering mechanical performance, research at the University of Cambridge [9] was conducted on the synthesis of optimal grid structures by considering multiple load cases. In another research [10], the main stress trajectories which are the representations of force flows on a free-form surface were used as a grid generation tool. However, the drawback of these two methods is that non-uniform grid with distorted unit cells came into being.

In practice, it is sometimes more important to generate grids on a curved surface that satisfy uniform criteria, where unit cells are with regular shapes and composed of fluent lines. Some research concentrated on the mesh generation over a surface without taking into account their structural performance. Muylle et al. [11] did research on mesh generation for finite element analysis such as presenting a new point creation scheme for generating unstructured, uniform-sized two-dimensional triangular meshes using the delaunay triangulation method. Pottman [12] produced grids manually for Yas Abu Dhabi Hotel roof relying on his experience in engineering. But this method is only appropriate for a specific project and there is a lack of versatility. Shepherd and Richens [6] proposed an interesting subdivision surface method for grid generation. An initial triangular or quadrilateral mesh was firstly imposed to a free-form surface, and then the mesh was subdivided for a number of iterations to fit the original surface. Zheleznyakova [13] proposed a new approach for triangular mesh generation based on the molecular dynamics method. Mesh nodes were considered as interacting particles. The node placement was determined by molecular dynamics simulation. Finally, well-shaped triangles were created after connecting the nodes by delaunay triangulation. Zheleznyakova [14] applied the above method to mesh on NURBS surfaces. The mesh was generated in the parametric domain of the NURBS surface patch using molecular dynamics simulation. Then, the well-shaped triangles were mapped from parametric space to 3D physical space. But the system of the interacting particles had a high sensitivity to the model parameter variation, leading to the problems of unstable solutions and slow convergences.

This paper is therefore innovative by combining surface flattening technique with guide line method to generate grids on a given free-form surface. Surface flattening enables the grid generation on a plane and solves the difficult problem of meshing a complex surface directly. Generating grids on the plane based on guide line method not only shows the connotation of architecture, but also controls the trends of free-form surface grids. Final examples show that this method is able to generate grids with excellent quality and therefore has preferable applicability in the structural concept design stage.

2 Surface representation

This paper expresses free-form surface by using NURBS (Non-uniform rational B-Splines) expression[15]. NURBS has been the industry standard that is used for shape representation, design and data exchange when geometric information is processed by computer. As a result, NURBS is a powerful tool in standard geometric design. NURBS realizes the arbitrary shape of surface by adjusting its control points, knots weights and establishes a one-to-one mapping relation between surface and parametric domain which is convenient for surface flattening and grid generation.

A *p*th-degree NURBS curve is shown in Fig.3 and is defined by [15]:

$$C(u) = \frac{\sum_{i=0}^{n} N_{i,p}(u) w_i CP_i}{\sum_{i=0}^{n} N_{i,p}(u) w_i} \quad a \le u \le b \tag{1}$$

where $\{CP_i\}$ are the control points, $\{w_i\}$ are the weights, and $\{N_{i,p}(u)\}$ are the *p*th-degree B-spline basis functions which are:

$$N_{i,0}(u) = \begin{cases} 1 \ \text{if} \ u_i \le u \le u_{i+1} \\ 0 \ \text{otherwise} \end{cases}$$
$$N_{i,p}(u) = \frac{u - u_i}{u_{i+p} - u_i} N_{i,p-1}(u) + \frac{u_{i+p+1} - u}{u_{i+p+1} - u_{i+1}} N_{i+1,p-1}(u) \tag{2}$$

defined on the non-periodic and non-uniform knot vector:

$$U = \{\underbrace{a,...,a}_{p+1}, u_{p+1},...,u_{r-p-1}, \underbrace{b,...,b}_{p+1}\} \tag{3}$$

where $r = n + p + 1$.

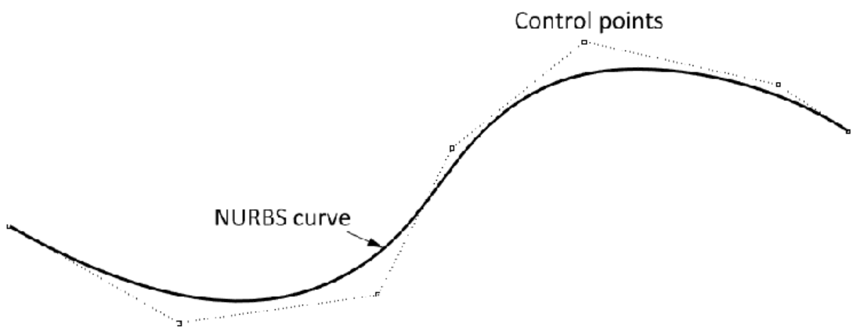


Fig.3 Relationship between a curve and its control points

An NURBS surface (shown in Fig.4) of degree *p* in the *u* direction and degree *q* in the *v* direction is a bivariate vector-valued piecewise rational function with the following form [15]:

$$S(u,v) = \frac{\sum_{i=0}^{n}\sum_{j=0}^{m} N_{i,p}(u) N_{j,q}(v) w_{i,j} CP_{i,j}}{\sum_{i=0}^{n}\sum_{j=0}^{m} N_{i,p}(u) N_{j,q}(v) w_{i,j}} \quad a \le u \le b, c \le v \le d \tag{4}$$

where $\{CP_{i,j}\}$ forms a bidirectional control net, $\{w_{i,j}\}$ are the weights, and $\{N_{i,p}(u)\}$, $\{N_{j,q}(v)\}$ are the non-rational B-spline basis functions defined on the knot vectors:

$$
\begin{cases}
U = \{\underbrace{a,\ldots,a}_{p+1}, u_{p+1},\ldots,u_{r-p-1}, \underbrace{b,\ldots,b}_{p+1} \\
V = \{\underbrace{c,\ldots,c}_{q+1}, v_{q+1},\ldots,v_{s-q-1}, \underbrace{d,\ldots,d}_{q+1}\}
\end{cases}
\tag{5}
$$

where $r = n + p + 1$ and $s = m + q + 1$.

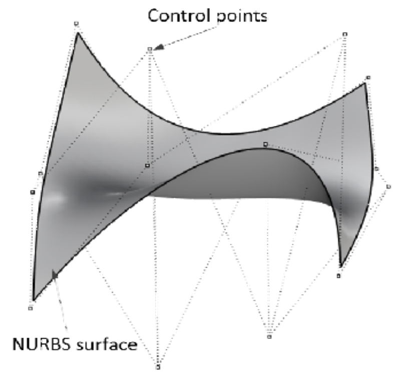


Fig.4 The relationship between a curved surface and its control points

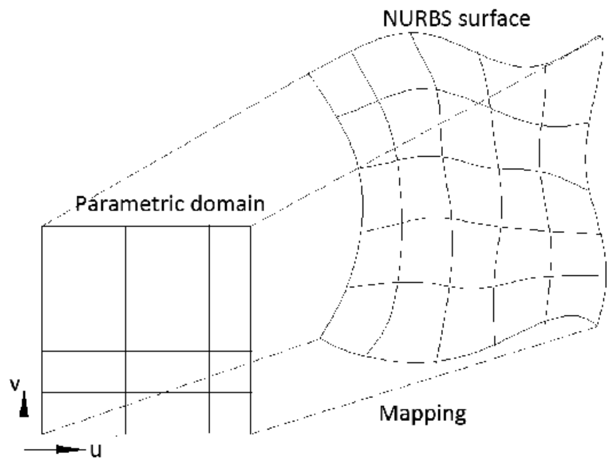


Fig.5 Mapping method from parametric domain to the NURBS surface

The curve mentioned above is defined in the three-dimensional space. However, in this article, the curves are always lying on the surface. It should be described by a two-time mapping rather than a single function, as shown in Fig.5. It is defined by an NURBS surface $S(u,v)$ and a two-dimensional NURBS curve $C(w)$ which lies on the parameter field of the surface. The value domain of the curve $C(w)$, is in the definitional domain of the surface $S(u,v)$. The pair of (u,v) is calculated from a parameter w by the function $C(w)$. The three-dimensional coordinate is subsequently calculated from the pair of parameters (u,v) by the function $S(u,v)$. It is obvious that all of the coordinates obtained by this two-time mapping method are exactly on the surface.

3 Surface Flattening

Surface flattening is widely used in engineering practice, particularly in the area of CAD/CAGD. For example, it is used to calculate and design blank shape in manufacture industry such as planes, cars, ships, furniture and clothing [16]. The Surface flattening technique has also been used in the generation of membrane cutting patterns. Topping and Iványi [17] introduced a method to transform the three dimensional strips of membrane surface to a plane using the dynamic relaxation method. In their method, the surface flattening process does not require the strip to be developable. According to the fact whether the surface is developable or not, surfaces are classified as developable surfaces and undevelopable surfaces. Gaussian curvature is used to define developability of surface and the sufficient/necessary condition for developable surface is that Gaussian curvature of surface is zero everywhere [18]. At present, there are three methods of surface flattening including geometric flattening, mechanical flattening and geometric flattening/mechanical amendment [16, 19, 20]. This paper will use geometric flattening method to unfold surface based on the rule of identical area. The NURBS surface in Fig.6 was selected as an example to explain the process of surface flattening.

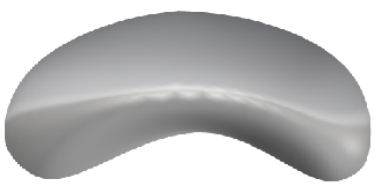
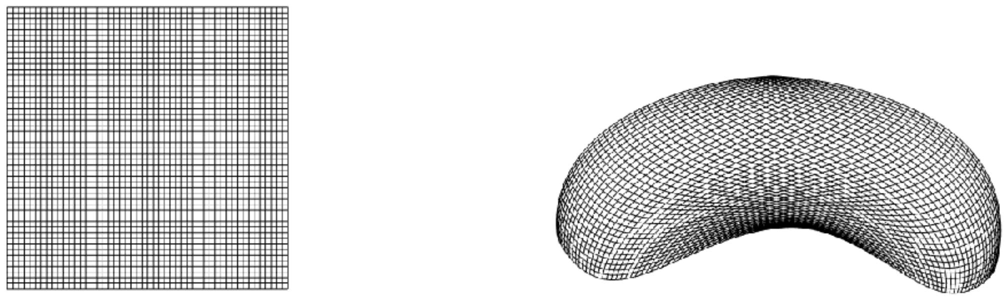


Fig.6 An NURBS surface

3.1 Surface discretizing

The parametric domain of the free-form surface is divided in *u* and *v* direction into a number of segments on the basis of relationship among surface adjacent boundary lengths. The parametric domain can be discretised into N×N or N×M grids corresponding to approximate equality or inequality of surface adjacent boundary lengths. Discrete free-form surface is generated by mapping dividing points of the parametric domain onto the surface (Fig.7).

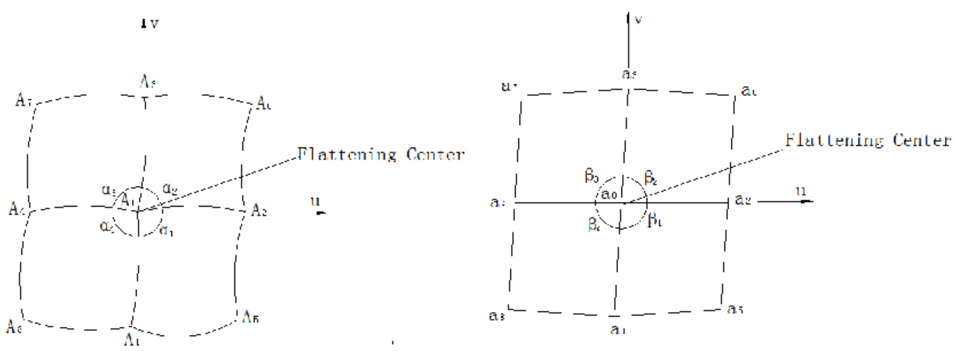


(a) Discrete mesh of the parametric domain (b) Discrete mesh of the surface

Fig.7 Surface discretising

3.2 Flattening of the central point and surrounding points

The central point $(u=1/2, v=1/2)$ of the surface grids is taken as the flattening center. The coordinates of unfolded plane points corresponding to the flattening center and its surrounding 8 points on the surface should be firstly determined, as shown in Fig. 8.



(a) Before flattening (b) After flattening

Fig.8 Flattening center and its surrounding points before and after flattening

The flattening center A_0 corresponds to point a_0 in the plane is regarded as origin of the flattening plane. Suppose γ is the summation of internal angles around the flattening center before and after flattening:

$$\gamma = 2\pi - (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) \tag{6}$$

γ will be allocated to 4 regions so that the corresponding angles on the plane after flattening can be obtained. After the flattening:

$$\beta_i = \alpha_i + \gamma \times \alpha_i \Big/ \sum \alpha_i \tag{7}$$

Suppose u and v directions have the same rate of expansion that is defined as t before and after flattening, which is shown as the following equation:

$$t = \frac{|a_0a_1|}{|A_0A_1|} = \frac{|a_0a_2|}{|A_0A_2|} = \frac{|a_0a_3|}{|A_0A_3|} = \frac{|a_0a_4|}{|A_0A_4|} \tag{8}$$

According to the rule that the sum of triangle $\triangle a_0a_1a_2$, $\triangle a_0a_2a_3$, $\triangle a_0a_3a_4$, $\triangle a_0a_4a_1$ areas after flattening is equal to the sum of the corresponding triangle areas before flattening, the value of t is obtained through the following equation:

$$\begin{aligned} &|A_0A_1||A_0A_2|\sin\alpha_1 + |A_0A_2||A_0A_3|\sin\alpha_2 + |A_0A_3||A_0A_4|\sin\alpha_3 + |A_0A_4||A_0A_1|\sin\alpha_4 \\ &= t^2(|A_0A_1||A_0A_2|\sin\beta_1 + |A_0A_2||A_0A_3|\sin\beta_2 + |A_0A_3||A_0A_4|\sin\beta_3 + |A_0A_4||A_0A_1|\sin\beta_4) \end{aligned} \tag{9}$$

Substituting t into equation (8), $|a_0a_1|$, $|a_0a_2|$, $|a_0a_3|$ and $|a_0a_4|$ are therefore obtained. Combined with formula (7), the coordinates of points a_1, a_2, a_3, a_4 can be achieved. Finally, according to the principle of identical area, the coordinates of points a_5, a_6, a_7, a_8 can be calculated. Taking a_5 as an example, suppose E_1, E_2, E_3 represent the area of triangle $\triangle A_1A_2A_5$, $\triangle A_0A_1A_5$, $\triangle A_0A_2A_5$ individually. Likewise, S_1, S_2, S_3 represent the area of the corresponding triangle after flattening. If there are three points: $P_1(x_1, y_1)$, $P_2(x_2, y_2)$, $P(x, y)$ to be in an anticlockwise arrangement in any plane. P_1, P_2 are fixed points and P is a moving point. The area of the triangle made up of these three points is as follows:

$$\varepsilon = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x & y & 1 \end{vmatrix} \tag{10}$$

Assume ΔS represents the summation of the squares of variation of the quadrilateral area before and after flattening:

$$\Delta S = \sum_{i=1}^{3}(S_i - E_i)^2 \tag{11}$$

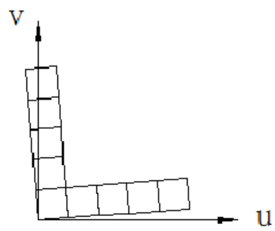
In order to get the minimum value of ΔS, the following formula is applied:

$$\begin{aligned} \partial \Delta S / \partial x = 0 \\ \partial \Delta S / \partial y = 0 \end{aligned} \tag{12}$$

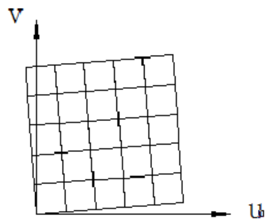
The x and y values which are the coordinates of point a_5 will then be obtained by solving the equations (12). a_6, a_7, a_8 coordinates can be obtained in the same way.

3.3 Grids flattening of the whole regions

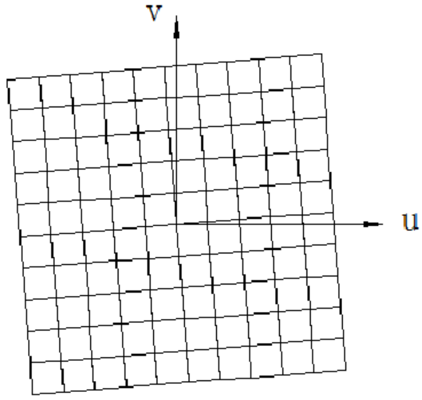
The entire surface is divided into four regions when the surface flattening center is determined, then the regions will be flattened. Following the flattening of A_0 and the surrounding 8 points, point A_2 which is on the right of point A_0 (Fig.8) is used as the center to flatten its surrounding points according to the principle of identical area. All grids will be flattened by repeating the above steps. Flattening steps are shown in Fig.9.



(a) Grid nodes of the first region flattening in a row



(b) Grid nodes of the first region flattening



(c) Grid nodes of four regions flattening

Fig.9 The steps of grids flattening

The discrete surface in Fig.7 is flattened based on above method, with the flattening result shown in Fig.10.

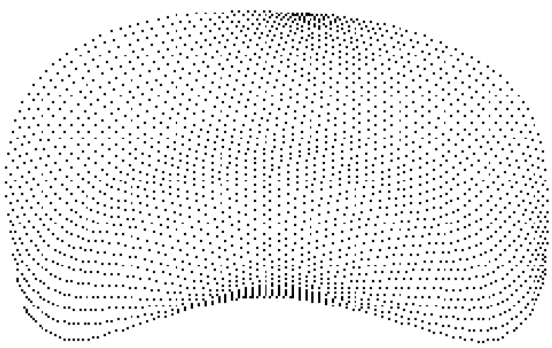


Fig.10 Rectangular lattices of the unfolded plane

This method takes into account of a number of factors such as the shape of surface and the relationship between surface adjacent boundary lengths when discretising the surface. The surface can be divided into N×M grids so that the method is also appropriate for narrow surface flattening. Geometric flattening method based on the principle of identical area can avoid generating cracks and overlaps during the surface flattening process. In addition, this method is easier to implement and is more adaptive in the current case.

4 Plane mesh generation based on guide line method

Following the flattening of the free-form surface, the flattened rectangular lattices are then fitted into the 2D surface [15] where grids are generated by the guide line method. The fitted 2D surface is shown in Fig.11. Free-form surface grid generation method based on guide line is in such a way that a curve which is sketched on the surface by an architect determines the grid trend. The basic idea of grid generation is that the entire surface will be covered with a number of guide lines by advancing the first guide line according to the features of surface. Then each guide line is divided into several segments and the dividing points between adjacent guide lines are connected into quadrilateral grids or triangular grids. Grid style of the surface has been largely determined by advancing the guide line so that guide line advancing is the key of surface meshing. According to the ways of guide line advancing, free-form surface meshing method based on guide line is classified into four types of translation method, offset method, scale method and two guide lines with two end vertices method. The translation method is chosen to generate grids on the NURBS surface in Fig.11 and detailed steps will be described in Sections 4.1-4.3.

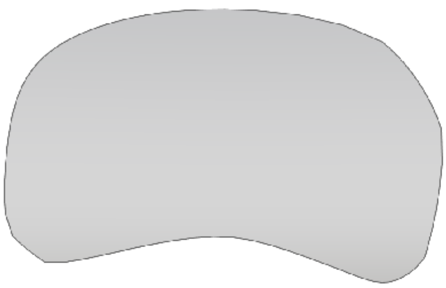


Fig.11 The fitted 2D surface

4.1 The guide line representation

As described in Section 2, the points on the NURBS surface and points on the parametric domain of the surface are in a one-to-one mapping relationship. Therefore, each curve on the NURBS surface and its corresponding curve on the parametric domain of the surface are also in a one-to-one mapping relationship.

Suppose that the NURBS expression of the projection curve on the parametric domain which corresponds to a curve in the 3D space, called C, is given:

$$\{u,v\} = C(t) = \frac{\sum_{i=0}^{n} N_{i,p}(t)\omega_i P_i}{\sum_{i=0}^{n} N_{i,p}(t)\omega_i}, \ a \le t \le b \tag{13}$$

where $\{C(t)\}$ is the standard expression of NURBS curve, $\{P_i\}$ are the control points, $\{w_i\}$ are the weights, and $\{N_{i,p}(t)\}$ are the *p*th-degree B-spline basis functions, as illustrated in Section 1. If all points of the projection curve on the parametric domain are mapped onto the surface, the expression of curve called C on the surface will be obtained:

$$\{x,y,z\} = D(u,v) = D(u(t),v(t)) \tag{14}$$

where $\{D(u,v)\}$ is the standard expression of NURBS curve, similar to $\{C(t)\}$. Eq. (14) means that if the expression of the projection curve on the parametric domain is known, the expression of corresponding curve on the surface will be achieved, accordingly. The projection curves on the parametric domain are thereby used to represent the corresponding curves on the surface.

Similarly, the first guide line, which is presented by the architect on the surface, can also be expressed by its projection curve on the parametric domain. If the subsequent translated guide lines are again expressed by their projection curves on the parametric domain, all guide lines will lie on the 3D surface.

The first guide line is sketched by the architect on the surface. In order to translate guide lines on the unfolded plane, the first guide line on the original surface should be projected to the unfolded plane primarily. Taking the NURBS surface in Fig.6 as an example, the first guide line on the surface and plane is shown in Fig 12 and Fig.13.

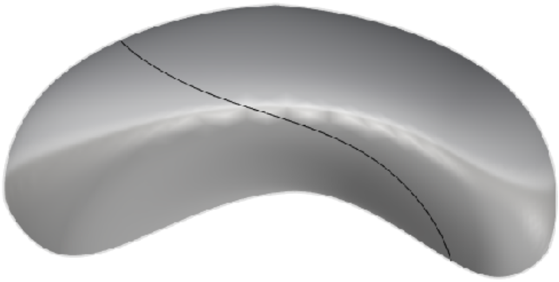


Fig.12 The guide line of surface

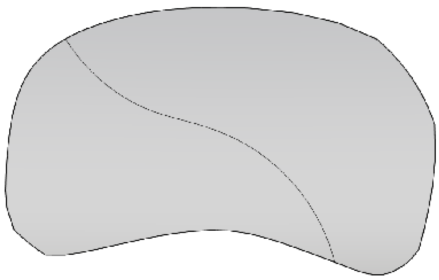


Fig.13 The guide line of unfolded plane

4.2 The guide line translation

Translation of guide line is in a way that a new guide line will be generated by moving all the points of the preceding guide line in a fixed length along the same direction. Suppose L is the optimum rod length, which is generally determined from the experience of the designer. If triangular mesh is to be guaranteed, the translation distance for each point of the preceding guide line will be $\frac{\sqrt{3}}{2}L$ in order to retain an equilateral triangle with side length L between two adjacent guide lines. However, for quadrilateral grid cell, the translation distance should be L so as to accommodate a square with side length L between two adjacent guide lines.

In general, the translation direction is changing with the translation of the guide lines. Consequently, the direction for each translation should be calculated according to geometric characteristics of the surface and the guide line. It is known that the translation direction of a guide line must be perpendicular with the guide line itself and the translation path is generally on the tangent plane of the surface. Suppose $\vec{D}$ is the translation direction of a guide line:

$$\vec{D} = \pm\vec{N} \times \vec{\tau} \tag{15}$$

where $\vec{N}$ is the normal vector of the surface, $\vec{\tau}$ is the tangent vector of the guide line, and $\pm$ represents two opposite translation directions when the guide line is in the middle of the surface. The translation direction will be calculated based on the midpoint of the guide line.

$$\begin{cases} \vec{D} = \pm\vec{N}\big(D\big(u(t), v(t)\big)\big) \times \vec{\tau}\big(D(t)\big) \\ t = (a+b)/2 \end{cases} \tag{16}$$

The midpoint of the guide line is the point when parameter t of the guide line is assigned with its intermediate value. The values of a and b are the minimum and maximum value of the knot vectors, respectively. And $D\big(u(t), v(t)\big), D(t)$ are the standard expressions of NURBS curve, $\vec{\tau}$ is obtained with the following formula:

$$\frac{dD(u(t),v(t))}{dt} = \frac{\partial D(u,v)}{\partial u} \times \frac{du}{dt} + \frac{\partial D(u,v)}{\partial v} \times \frac{dv}{dt} \tag{17}$$

The next stage is to translate the guide line. It is impracticable to translate every point of the guide line because there are infinite points in a curve. A number of approximate equidistant points of the first guide line are therefore selected and translated a fixed distance in the same direction. Then the points after translation are projected onto the surface. Following, the points on the surface should be projected onto the parametric domain to fit these points into an NURBS curve. This NURBS curve on the parametric domain must correspond to a curve on the surface. The corresponding curve which is not only on the surface but also passing through the translated approximate equidistant points is the new guide line. Finally, the above steps will be repeated to obtain a surface that is covered with guide lines. Taking the NURBS surface in Fig.13 as an example, the resulted plane covered with guide lines is shown in Fig.14.

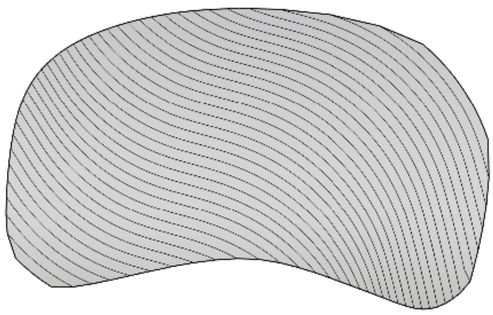


Fig.14 The plane covered with guide lines

One of the two projection processes mentioned above is to project a number of approximate equidistant points of the guide line to the surface, and the other is to project points in the surface to the parametric domain. These processes are called point inversion or projection for surfaces and their algorithms are as follows. Given a point $P=(x,y,z)$, assumed that the point lies in the NURBS surface $S(u,v)$, point inversion is to find the corresponding parameter $(\bar{u},\bar{v})$, such that $S(\bar{u},\bar{v})=P$. Form the vector function:

$$r(u,v)=S(u,v)-P \tag{18}$$

and the two scalar equations

$$\begin{cases} f(u,v)=r(u,v)\cdot S_u(u,v)=0 \\ g(u,v)=r(u,v)\cdot S_v(u,v)=0 \end{cases} \tag{19}$$

To find the solution to problem defined by Eq. (19). Let

$$\delta_i = \begin{bmatrix} \Delta u \\ \Delta v \end{bmatrix} = \begin{bmatrix} u_{i+1} - u_i \\ v_{i+1} - v_i \end{bmatrix}$$

$$J_i = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix} = \begin{bmatrix} |S_u|^2 + r \cdot S_{uu} & S_u \cdot S_v + r \cdot S_{uv} \\ S_u \cdot S_v + r \cdot S_{vu} & |S_v|^2 + r \cdot S_{vv} \end{bmatrix}$$

$$\kappa_i = -\begin{bmatrix} f(u_i, v_i) \\ g(u_i, v_i) \end{bmatrix}$$

where all the functions in the matrix J_i are evaluated at (u_i, v_i). At the *i*th iteration, the 2 x 2 system of linear equations should be solved to obtain δ_i, given by

$$J_i \delta_i = \kappa_i \tag{20}$$

From δ_i,

$$u_{i+1} = u_i + \Delta u, \quad v_{i+1} = v_i + \Delta v \tag{21}$$

Convergence criteria are given by

$$\left|(u_{i+1} - u_i) S_u(u_i, v_i) + (v_{i+1} - v_i) S_v(u_i, v_i)\right| \le \varepsilon_1$$

$$\left|S(u_i, v_i) - P\right| \le \varepsilon_1$$

$$\frac{\left|S_u(u_i, v_i) \cdot (S(u_i, v_i) - P)\right|}{\left|S_u(u_i, v_i)\right| \left|S(u_i, v_i) - P\right|} \le \varepsilon_2, \quad \frac{\left|S_v(u_i, v_i) \cdot (S(u_i, v_i) - P)\right|}{\left|S_v(u_i, v_i)\right| \left|S(u_i, v_i) - P\right|} \le \varepsilon_2 \tag{22}$$

Again, the conditions are checked by

1. Point coincidence:

$$\left|S(u_i, v_i) - P\right| \le \varepsilon_1 \tag{23}$$

2. Zero cosine:

$$\frac{\left|S_u(u_i, v_i) \cdot (S(u_i, v_i) - P)\right|}{\left|S_u(u_i, v_i)\right| \left|S(u_i, v_i) - P\right|} \le \varepsilon_2, \quad \frac{\left|S_v(u_i, v_i) \cdot (S(u_i, v_i) - P)\right|}{\left|S_v(u_i, v_i)\right| \left|S(u_i, v_i) - P\right|} \le \varepsilon_2 \tag{24}$$

If these conditions are not satisfied, a new value u_{i+1} is computed using Eq. (21). Then two more conditions are checked:

3. Ensuring that the parameters stay in range $u_{i+1} \in [0,1], v_{i+1} \in [0,1]$:

If the surface is not closed in the *u* direction:

$$\text{if} \quad (u_{i+1} < 0), \; u_{i+1} = 0$$

$$\text{if} \quad (u_{i+1} > 1), \; u_{i+1} = 1$$

If the surface is not closed in the *v* direction:

$$\text{if} \quad (v_{i+1} < 0), \; v_{i+1} = 0$$

$$\text{if} \quad (v_{i+1} > 1), \; v_{i+1} = 1$$

If the surface is closed in the *u* direction:

$$\text{if} \quad (u_{i+1} < 0), \; u_{i+1} = 1 - (0 - u_{i+1})$$

$$\text{if} \quad (u_{i+1} > 1), \; u_{i+1} = 0 + (u_{i+1} - 1)$$

If the surface is closed in the *v* direction:

$$\text{if} \quad (v_{i+1} < 0), \; v_{i+1} = 1 - (0 - v_{i+1})$$

$$\text{if} \quad (v_{i+1} > 1), \; v_{i+1} = 0 + (v_{i+1} - 1)$$

4. Parameters do not change significantly, therefore:

$$\left| (u_{i+1} - u_i) S_u(u_i, v_i) + (v_{i+1} - v_i) S_v(u_i, v_i) \right| \le \varepsilon_1$$

Iteration is halted if any of conditions 1, 2, or 4 is satisfied. u, v values obtained are the results of the point inversion or projection for surfaces [15].

4.3 The guide line segmentation and mesh generation

A number of guide lines can be generated by following the steps in Section 4.1 and 4.2, using the translation procedure. Every guide line on the surface should be approximately divided into a number of segments in order to form a series of dividing points according to the given optimum rod length. It should be noted that free-form surface is presented in the form of grids which are formed by rods. Suppose the rod length of the grid is the chord length of the curve. The chord length of every arc rather than the arc length itself will be calculated after the guide line division, as shown in Fig.15.

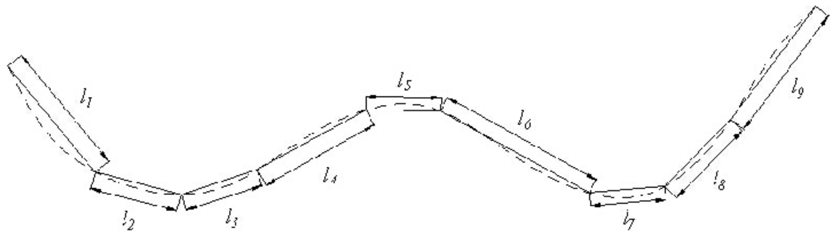


Fig.15 The guide line division

The dividing points of two adjacent guide lines are linked into straight lines which represent grid rods after guide line segmentation. The related division points in the guide lines are thereby, the joints of the single layer lattice shell. Taking the NURBS surface in Fig.14 as an example, the resulted grids are shown in Fig.16.

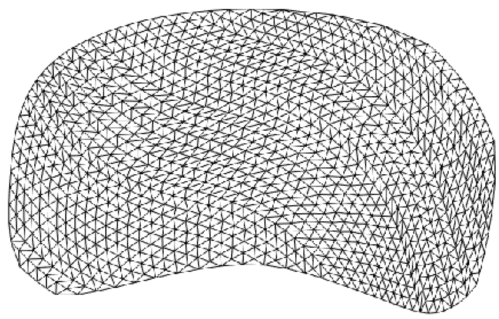


Fig.16 Unfolded plane mesh

As shown in Fig. 16, the grids generated by the method based on the guide line not only meet the requirements of regular shape and fluent lines, but also embody the trends of free-form surface grids.

5 Space mapping of plane grids

After generating grids on the unfolded plane, the plane grids need to be mapped onto the space surface. The mapping relationship among unfolded plane, parametric domain and space surface should be clarified before mapping. There is a one-to-one corresponding relationship among the points of unfolded plane, parametric domain and space surface. A procedure to map the plane grids into space surface is as follows:

Step1 The parameter domain coordinates (u,v) of unfolded plane's grid nodes should be obtained by projecting all grid nodes to the parametric domain. Detailed algorithms have been introduced as point inversion or projection for surfaces in Section 4.3.

Step2 The *u* and *v* values obtained from Step1 are substituted into the NURBS equation of space surface to obtain coordinates of grid nodes on the surface. Detailed algorithms are as follows. Five steps are required to compute a point on a surface at fixed (u,v) parameter values [15]:

(1) Find the knot span in which *u* lies by a linear search method, suppose $u \in [u_i, u_{i+1})$;

(2) Compute the nonzero basis functions $N_{i-p,p}(u), \cdots, N_{i,p}(u)$ by the deBoor recurrence formula;

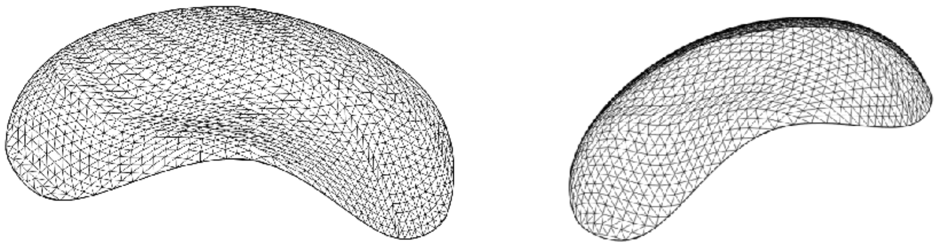
(3) Find the knot span in which *v* lies by a linear search method, suppose $v \in [v_j, v_{j+1})$;

(4) Compute the nonzero basis functions $N_{j-q,q}(v), \cdots, N_{j,q}(v)$ by the deBoor recurrence formula;

(5) Substitute the nonzero basis functions, corresponding control points and weights into the formula (4) to get the three-dimensional coordinates of the surface point.

Step3 Suppose the grid topology is unchanged before and after mapping, the grid nodes of surface are linked into grids according to the topological relation of the plane grids.

Taking the surface in Fig.12 as an example, the final spatial grids are shown in Fig.17.



(a) Top (b) Perspective

Fig.17 Space mesh of surface

The framework defined above has been developed in the operating system of Microsoft Windows, using C++ language. A number of examples of grid generation on free form surface will be used to illustrate the successful execution of the methodology.

6. Grid quality evaluation

In the process of grid generation, the quality of the mesh can be evaluated through a visual check on screen. However, this might not be a sensible choice for the application in large scale structures. A grid quality criterion is therefore necessary in order to provide a quantitative evaluation on the generated grid. The uniformity of grid is thereby defined to ensure a grid pattern which is of uniform rod length and regular cell to meet aesthetic requirements. This is due to the fact that the regular cell shape is the precondition for the fluency of while rod length uniformity facilitates the construction and manufacturing process.

This paper therefore uses a grid uniformity quality index which is able to evaluate the shape and rod length quality. The shape quality reflects regularity of generated grid cell while the rod length quality is used to ensure uniform size of the grids. A brief introduction on the evaluation of shape and rod length quality is first given and then a comprehensive comparison of different grid generation methods with and without using the surface flattening method is provided.

6.1 Shape qu ality

For the triangular mesh, shape quality of the equilateral triangle mesh is the best. For an arbitrary triangle mesh unit ΔABC whose area is $S_{\Delta ABC}$, assuming that AB, BC, AC represent the three triangle side lengths. The shape quality indicator of this triangle which is called α shown as following:

$$\alpha = 4\sqrt{3}\frac{S_{\Delta ABC}}{AB^2 + BC^2 + AC^2} \tag{25}$$

where the value of α is between 0-1. Higher value of α indicates a better shape quality of a grid cell, and when the value of α is 1, the triangle is equilateral triangle whose shape quality is considered to be the best.

Likewise, ▱ABCD represents the quadrilateral grid cell with four side lengths: AB, BC, CD and AD. The shape quality indicator of this quadrilateral which is called β, which is defined as:

$$\beta = 4\sqrt[4]{\frac{S_{VABC}\ ?\ S_{VBCD}\ ?\ S_{VCDA}\ ?\ S_{VABD}}{\left(AB^2 + AD^2\right)?\left(AB^2 + BC^2\right)?\left(BC^2 + CD^2\right)?\left(CD^2 + AD^2\right)}} \tag{26}$$

where the value of β is between 0-1, again, higher value of β indicates better shape quality of quadrilateral grid cell.

In order to evaluate shape quality of the whole grid structure, the shape quality index of the composed grid cells are regarded as a sample. N, n_1, n_2 represent the total number, the number of triangle and the number of quadrilateral grid cells respectively. The average value $\bar{E}$ and the standard deviation δ_E of shape quality index of the whole mesh are thereby defined as:

$$\bar{E} = \frac{1}{N}\left(\sum_{i=1}^{n_1}\alpha_i + \sum_{j=1}^{n_2}\beta_j\right) \tag{27}$$

$$\delta_E = \sqrt{\frac{\sum_{i=1}^{n_1}\left(\alpha_i - \bar{E}\right)^2 + \sum_{j=1}^{n_2}\left(\beta_j - \bar{E}\right)^2}{N-1}} \tag{28}$$

Higher value of $\bar{E}$ indicates better shape quality of the generated grid structure while minor value of δ_E has smaller variation of grid cell shapes, which means a more uniform distribution of the grids.

6.2 Rod length quality

Rod is the basic unit of grid structure, which means uniform rod lengths are critical for aesthetics and fluency of mesh. Besides, uniform length rods will better satisfy the constructional and manufacturing constraints. In order to evaluate the rod length quality of the generated grid structure, the lengths of all grid rods are taken as a sample whose size is M. The average value $\bar{L}$ and the standard deviation δ_L of rod lengths are defined by:

$$\bar{L} = \frac{1}{M}\sum_{i=1}^{M} L_i \tag{29}$$

$$\delta_L = \sqrt{\frac{\sum_{i=1}^{M}\left(L_i - \bar{L}\right)^2}{M-1}} \tag{30}$$

where smaller value of δ_L has more uniform grid rod lengths.

7. Example I

7.1 Comparison of different guide lines

Taking the NURBS surface in Fig.6 as an example, the approximate horizontal curve (Fig. 18) and the approximate vertical curve (Fig. 20) were chosen to be guide lines of the surface. By using the methodology proposed in this paper, the resulted grids are shown in Figs. 19 and

21.

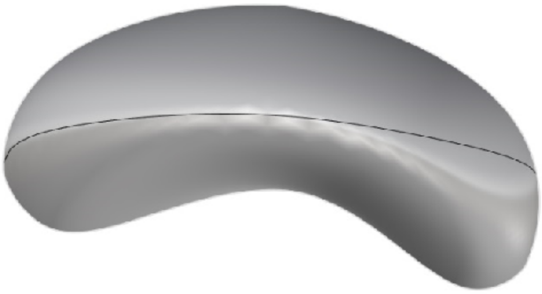
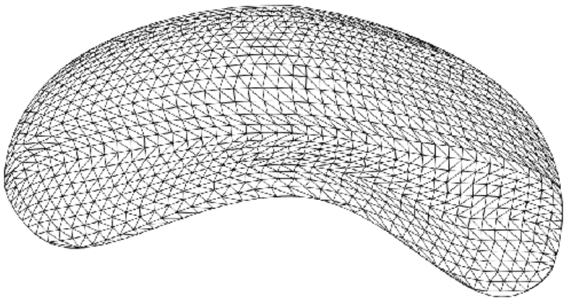
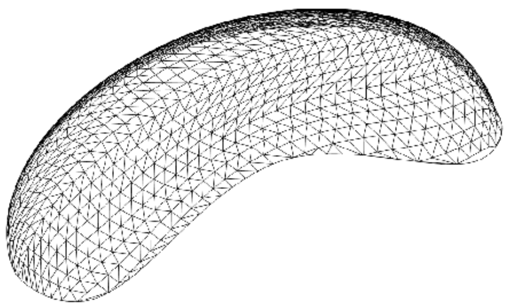


Fig.18 The horizontal guide line of surface



(a) Top



(b) Perspective

Fig.19 Surface mesh based on horizontal guide line

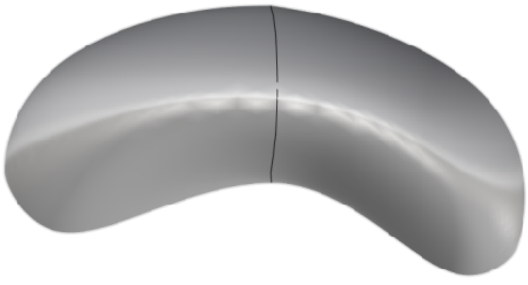
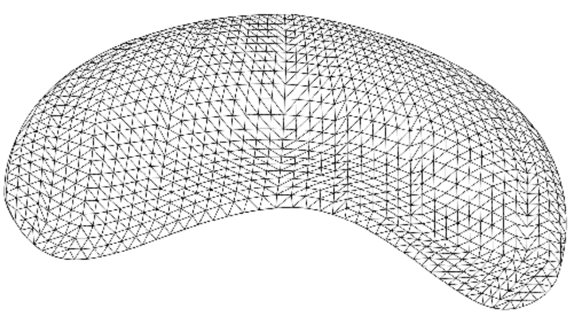
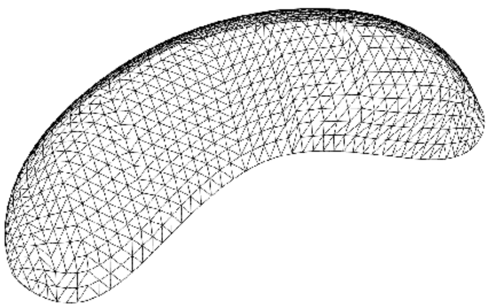


Fig.20 The vertical guide line of surface



(a) Top



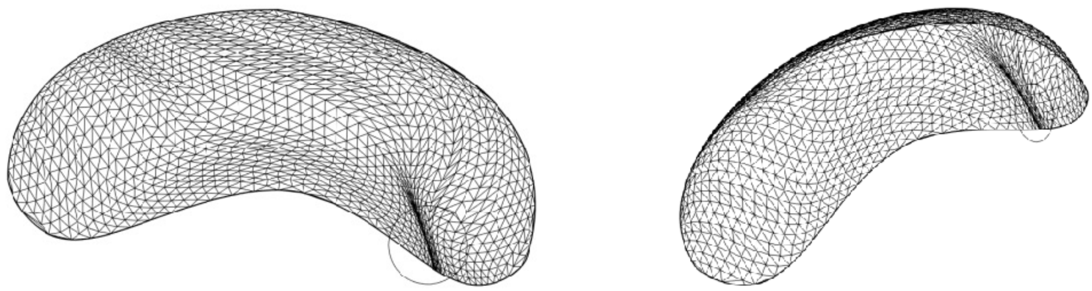
(b) Perspective

Fig.21 Surface mesh based on vertical guide line

The above results show that grids generated by surface flattening and guide line advancing based on different guide lines are of good quality, regular shape and with fluent lines. In addition, by comparing the results shown in Fig.19 with Fig.21, it is shown that the guide lines not only control the trends of free-form surface grids, embody the connotation of architecture, but also have a great influence on the quality of the final grid. The diagonal guide line can generate a more optimised mesh among three guide lines chosen above.

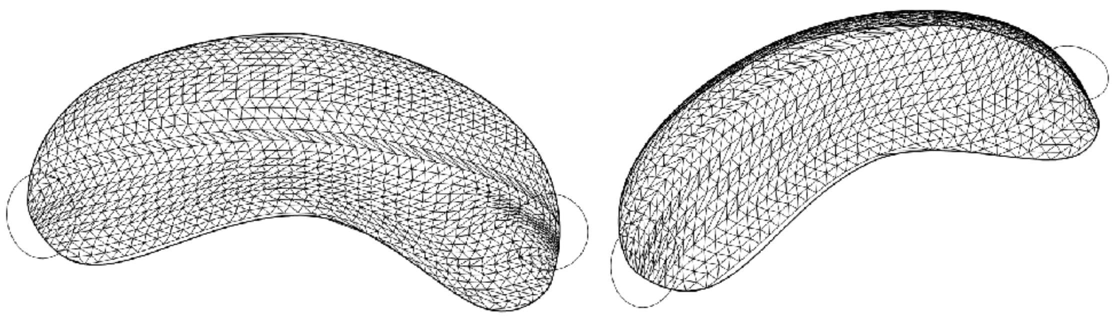
7.2 Comparison of grid generation methods

In this Section, the results from guide line method with and without surface flattening will be compared. Diagonal guide line, horizontal guide line and vertical guide line are still chosen to generate grids on the surface in Fig.6. The resulted grids based on just the guide line method without surface flattening are shown as follows.



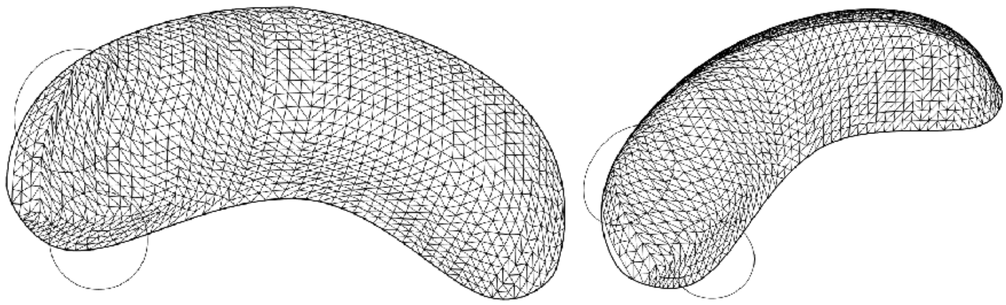
(a) Top (b) Perspective

Fig.22 Surface mesh based on diagonal guide line



(a) Top (b) Perspective

Fig.23 Surface mesh based on horizontal guide line



(a) Top (b) Perspective

Fig.24 Surface mesh based on vertical guide line

The above results show that the quality of mesh generated by only guide line method is excellent in general compared with Fig.19, 21 which are based on both surface flattening technique and guide line method. But in the area with sharp curvature, the problems of guide lines stacking, grids overlapping, and unsmooth grid happened, as shown in the circles of Fig.22-24. Since advancing the guide line directly on the surface with fixed advancing distance will cause non-uniform guide lines when the curvature varies greatly. The planar curvature is zero and relatively uniform guide lines are accordingly obtained.

The standard deviations of shape quality index and rod length index of the surface mesh described in Section 6 are therefore presented in Table 1 and Table 2. It is shown in the tables that the grids generated by combining surface flattening and guide line advancing method are of better quality. The grid shape quality index and the deviation of rod length of the grid structure are reduced by up to 47% and 34% respectively by using the guide line method with surface flattening when compared to the method without surface flattening.

Table 1. The standard deviations of the grid shape quality index of the grid structure

Grid generation methods	Diagonal guide line	Horizontal guide line	Vertical guide line
With surface flattening (δ_{E1})	0.183	0.236	0.203
Without surface flattening (δ_{E2})	0.332	0.406	0.384
$\frac{\delta_{E1}-\delta_{E2}}{\delta_{E2}}\times 100\%$	-44.88%	-41.87%	-47.14%

Table 2. The standard deviations of rod length of the grid structure

Grid generation methods	Diagonal guide line	Horizontal guide line	Vertical guide line
With surface flattening (δ_{L1})	0.426	0.466	0.435
Without surface flattening (δ_{L2})	0.621	0.687	0.663
$\frac{\delta_{L1}-\delta_{L2}}{\delta_{L2}}\times 100\%$	-31.40%	-32.17%	-34.39%

7.3 Quadrilateral mesh

In the previous examples, starting from a predefined surface, a grid structure with triangular cells was generated while in this Section, 2 initial guide lines will be selected and quadrilateral grid mesh will be generated over the curved surface. Taking two intersected curves as initial guide lines, by combining the surface flattening technique with guide line method, the generated final results are shown in Fig.27. It is shown that a grid patter generated is of good quality, regular shape and with fluent lines.

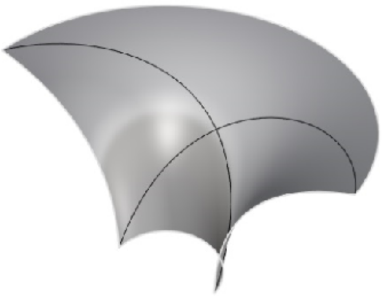


Fig.25 NURBS surface and the guide lines

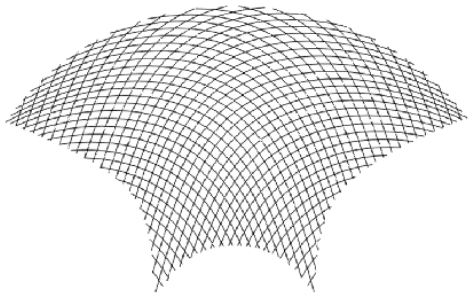
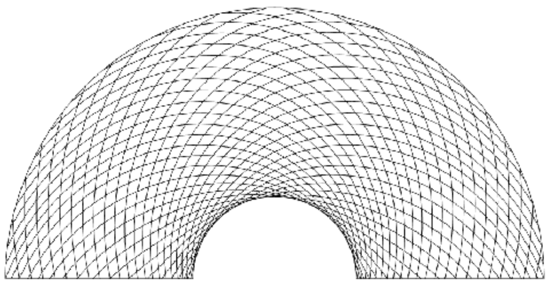
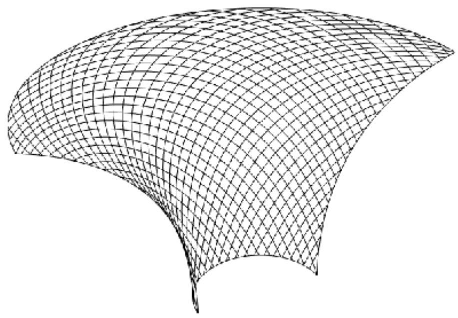


Fig.26 Unfold plane mesh



（a）Top



（b）Perspective

Fig.27 Surface mesh

In Section 7.4, 3 more examples are presented by using the proposed guide line method with surface flattening, as shown in Fig.28-Fig.33.

7.4 Example II

7.4.1 Example 1

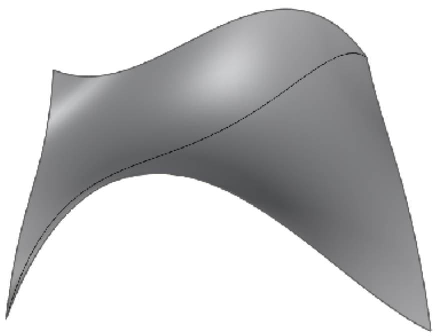
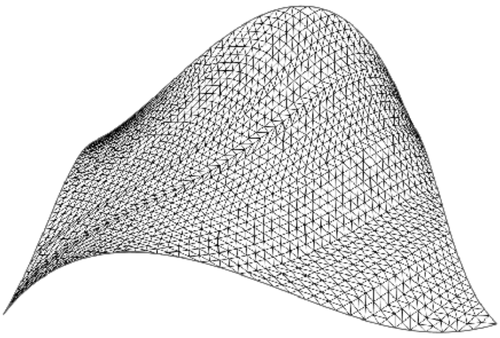
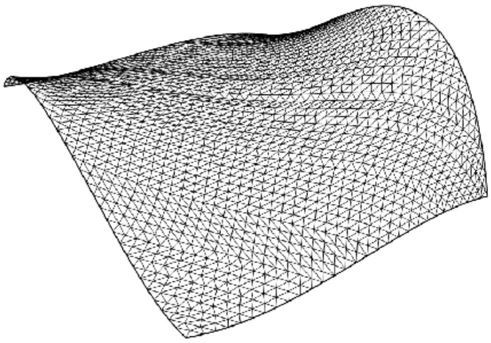


Fig.28 Surface with the initial guide line





（a） Perspective （b） Perspective

Fig.29 Surface mesh

7.4.2 Example 2

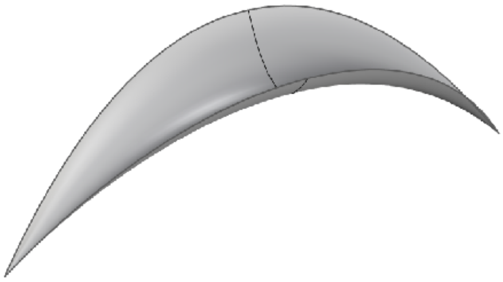
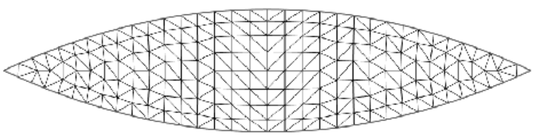
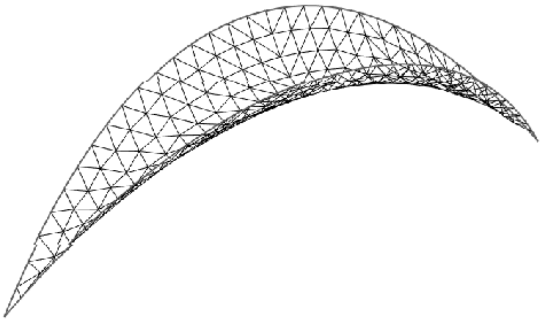


Fig.30 Surface with initial guide line





（a） Top （b） Perspective

Fig.31 Surface mesh

7.4.3 Example 3

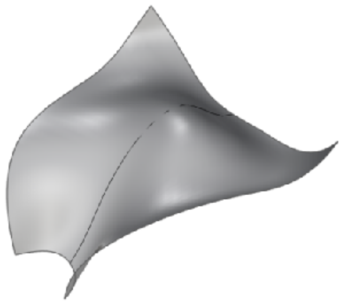
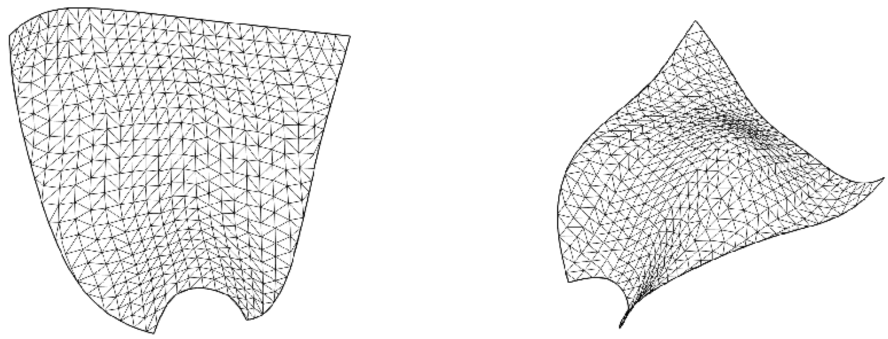


Fig.32 Surface with the guide line



（a） Top （b） Perspective

Fig.33 Surface mesh

8. Comparison the guide line methods with and without surface flattening

This paper proposed a grid generation methodology on free-form surface based on surface flattening and the guide line grid generation method. With an additional surface flattening procedure, a grid pattern with higher quality can be achieved (see Section 6) using the guide line method. It has been pointed out in the paper that the guide line translation is the key process of the guide line method. The guide line translation includes three factors: the translation distance, the translation direction and the translation methodology.

Among these three factors, the translation direction is changing with the translation of the guide lines. Consequently, the direction for each translation should be calculated according to geometric characteristics of the surface and the definition of initial guide line. It is known in Section 6 that the translation direction of a guide line must be perpendicular with the guide line itself and the translation path is on the tangent plane of the surface. The curvature of the surface therefore has an effect on the translation direction. The guide lines on the unfolded plane after translation can be more uniform since the curvature of plane is zero everywhere. However, when subjected to severe curvature variation, a problem of guide line stacking or distortion will arise by using the guide line advancing method without the surface flattening. Therefore, the guide line method with surface flattening is more advanced when dealing with the grid generation on complex surfaces.

9 Conclusions

This paper puts forward a grid generation technique on free-form surface for lattice structure design use, based on surface flattening technique and the guide line grid generation method. The first step of this method was to flatten the surface into a plane. The surface were divided into N×M grids by taking factors such as shape of surface and relationship between adjacent surface boundary lengths into account before the flattening process. The free-form surface was then flattened based on the principle of identical area. A program based on C++ was developed in order to facilitate the grid generation process. The results show that the grid shape quality index and the deviation of rod length of the grid structure are reduced by up to 47% and 34%, respectively, using the guide line method with surface flattening when compared to the method without surface flattening. The generated grids based on the guide line method not only met the requirements of regular shape and fluent lines, but also embodied the trends of grids and the connotation of architecture.

Acknowledgement

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